

Handwritten table with 11 columns and 2 rows. The first row is labeled 'si' and the second row is labeled 'fi'. The first row contains values 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11. The second row contains values 1, 3, 0, 5, 3, 5, 6, 8, 8, 2, 12. Below the first row, there are circled numbers 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14. Above the first row, there are arrows indicating a sequence of operations: 1 down arrow, 2, 3, 4 down arrow, 5, 6, 7, 8 down arrow, 9, 10, 11 down arrow.

si	1	2	3	4	5	6	7	8	9	10	11
fi	1	3	0	5	3	5	6	8	8	2	12
	4	5	6	7	8	9	10	11	12	13	14

Greedy choice (*)

a_1
 a_4
 a_8
 a_{11}

$O(n^3)$

$O(n)$ ✓

$n \log n + O(n)$ ✓

so that's why time complexity will be end log in plus order of n Which is